Hierarchical Learned Risk-Aware Planning for Modeling Human Driving Behavior in Virtual Environments

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***Abstract*—Accurately replicating human driver behavior in simulation is paramount for testing safety-critical systems of autonomous vehicles. Despite extensive research, a comprehensive method for modeling human driving behavior remains elusive. Current methodologies predominantly use imitation learning or inverse reinforcement learning to simulate human driving behavior. However, these techniques replicate human driving behavior exclusively in the specific contexts in which the models have been trained, and do not account for the differing styles of drivers. Recent work in risk-aware planning has shown effectiveness in replicating the human decision-making process. However, current implementations of risk-aware planners use naive risk predictions from surrounding vehicles, limiting the effective planning horizon time. This restriction confines risk-aware models to reactive maneuvers, and consequently, they cannot effectively model human drivers in simulation. This paper introduces a novel hierarchical risk-aware framework designed to model human drivers in simulation by integrating trajectory predictions of surrounding vehicles to produce accurate risk levels. We subsequently learn a risk-aware reward function from human agents that accounts for differing driving styles. This framework is specifically designed to generate new, human-like behavior of multiple driving styles in a variety of contexts, offering a new approach to simulating human drivers for safety-critical autonomous vehicle research.**

***Index Terms*—Risk-aware; Human modeling; Autonomous Vehicles; Simulation; Robotics**

I. Introduction

As the production of autonomous vehicles increases and their presence on our roads becomes more commonplace, thoroughly testing the safety algorithms within these vehicles becomes critical. This is a vital step towards safeguarding all road users. However, real-world testing of these systems presents several challenges. It is not only costly and time-consuming but also presents potential safety hazards, especially when testing scenarios involving near-collision events.

Given these drawbacks, simulations provide a compelling alternative to real-world testing. While real-world validation of systems remains essential, conducting a substantial part of the testing in a simulated environment offers numerous advantages. It is not only safer, more efficient, and cost-effective, but it also presents an opportunity to test and optimize the system under a variety of challenging scenarios without physical risk.

Nevertheless, one of the main complexities in simulation testing arises from the unpredictable nature of human drivers. Most near-collision situations involving autonomous vehicles occur due to unpredictable human behavior. Therefore, for simulation testing of autonomous vehicle systems to be accurate and effective, it is critical to incorporate a reasonably precise model of human driving behavior. This aspect is fundamental to ensure that the algorithms can effectively deal with real-world unpredictability and enhance overall road safety.

Although various approaches for modeling human behavior have been proposed, no accurate and comprehensive model of human driving behavior has been widely adopted. In order to achieve human-like behavior, the trajectory of the ego vehicle must be planned several seconds into the future. Many models limit the planning horizon of the trajectory to around 1 second [1], but these models fail to accurately replicate many human behaviors such as overtaking a vehicle due to being unable plan such maneuvers with their short planning horizons. An intuitive understanding of how human drivers operate their vehicles shows that human drivers think about the actions they will take when driving many seconds into the future and make decisions based on these long-term plans.

To provide an accurate model of human driving behavior in simulation, we propose a novel hierarchical learned risk-aware planning framework. Our framework uses a simple recurrent neural network-based trajectory prediction method to predict the actions of all surrounding vehicles, then uses those predicted trajectories to generate risk values along a highway that a path is planned through and then controlled to. We use NGSIM data to learn parameters that determine human driving and feed those into the risk-aware planning cost function to more accurately represent human driving. Since our proposed model is able to plan a trajectory 5 seconds into the future in less than a tenth of a second, this framework is effective at capturing high-level human behaviors, while still allowing it to be effectively used in simulation. Additionally, the model includes a parameter for allowable risk, allowing the model to be adjusted to represent different styles of driving, increasing the accuracy of the model in various situations.

Our **main contributions** are: **(1)** We present a novel risk-aware framework for generating new realistic human trajectories in simulated environments; **(2)** We offer an analysis of our framework in a highway driving scenario and validation of our model on the NGSIM dataset; **(3)** We provide examples of newly generated trajectories showing human behavior not found in other methods for modeling human drivers.

II. Related Work

Rule-based approaches attempt to model human driving behavior by intuiting how a human driver will maneuver their vehicle and defining a comprehensive set of rules to represent that behavior. These approaches often generalize well to a variety of situations, and can easily generate new behaviors not previously observed in datasets. These models also have the benefit of generating easily interpretable trajectories. There are many rule-based approaches for modeling human behaviors such as the Intelligent Driver Model (IDM) [2], Game-theoretic planning [3] [4], and risk-aware planning [5] [6]. The IDM works well in its context, but can only replicate lane following without lane changing making it unable to replicate human behavior. Other approaches on their own tend to fail to behave sufficiently similar to human drivers due to the complexities of human driving [7].

Learning-based approaches for modeling human driving behavior generally use Inverse Reinforcement Learning [1] [8] [9], or Imitation Learning [9] [10] [11] to learn cost functions for human drivers or to replicate expert driving maneuvers respectively. Both of these methods generally perform reasonably well at replicating human driving behavior when tested in situations already observed in data; however, both methods tend to fail to generalize well to new situations and contexts [12]. Additionally, learning-based methods require large amounts of data for training, which is not always available for many driving scenarios. Another problem of these learning-based models is that they are able to capture the specific maneuvers of individual vehicles but fail to learn high-level behaviors of vehicles [12].

Human driving behavior depends on a number of factors; however, it has been shown that driving behavior can be predicted with reasonable accuracy using a risk-based method [6] [13] [14] [15]. By assuming that drivers operate their vehicle primarily based on risk, and perceived optimality, we can develop an effective driver model. The NGSIM dataset provides valuable information on human driving behavior with proper filtering of data [16], [17], [18]. By using these data, we can observe human driver tendencies and behaviors to both learn models and validate them on real-world data. Our novel approach to modeling human driving behavior has distinct advantages over existing methods in multiple key areas. By predicting the trajectories of surrounding vehicles 5 seconds into the future, we enable the risk-aware planner to plan for many maneuvers that may be less optimal in the short term but yield long-term rewards, similar to how human drivers often operate their vehicles. Additionally by learning parameters for the risk-aware planning we can gain many of the benefits of both rule-based and learned models by having a well defined set of driving rules that are generalized to perform well in new road contexts and situations, while still accurately representing human driving behavior.

III. Preliminaries

A. Trajectory Prediction

[*Summarized from* [19]] Convolutional Social Pooling for trajectory prediction is the means of estimating a set of probable trajectories for a vehicle. The trajectory prediction is performed by a set of Long Short-Term Memory (LSTM) networks that are used to encode the histories of surrounding vehicles into a social occupancy grid. These histories are then passed through multiple convolutional and pooling layers before being decoded by another LSTM network into predicted trajectories. The predictor outputs six individual trajectories, representing three lateral maneuvers: left lane change, keep lane, and right lane change; and two longitudinal maneuvers: maintaining velocity or decelerating. Each of these trajectories is assigned an appropriate probability based on its likelihood of occurrence.

B. Risk-Aware Planning

[*Summarized from* [6]] Treating all objects and other agents in the system as dynamic obstacles in an environment where $Q \subset \mathbb{R}^N$ with all points $q \in Q$. Agent positions are denoted as p_i where $i \in 1, ..., n$, and the agents are modeled with double integrator dynamics such that $\ddot{p}_i = u_i$ where u_i is the control input for the ith agent. We assume that all agents have a max speed and acceleration such that $||\ddot{p}_i|| < a_i$ and $||\dot{p}_i|| < v_i$. The positions of all agents are denoted as $p = [p_1^T|...|p_n^T]^T$. To generate a rectangular higher-order Gaussian that equally weights the risk along a vehicle's entire width, the cost function may be computed by Equation 1.

$$\hat{\mathcal{H}}(q, p_i, \dot{p}_i) = \sum_{i=1}^{n} \frac{exp\left(-\left(\frac{(q_x - p_{x,i})^2}{\sigma_{x,i}^2}\right)^{\beta} - \left(\frac{(q_x - p_{y,i})^2}{\sigma_{y,i}^2}\right)^{\beta}\right)}{1 + exp(\alpha \dot{p}_i^T (q - p_i))} \tag{1}$$

IV. Method

A. Risk-aware Planner

Using the Convolutional Social Pooling trajectory prediction, we predict 5 seconds of future trajectories. We selected the trajectory prediction used in [19] due to the availability of the code and the dataset used for training. These trajectories consist of 25 individual estimated poses for the vehicles, each pose being temporally separated by 0.2 seconds. For each of the six predicted futures that have a probability greater than 10 percent, we generate a risk field. We found that setting a 10 percent threshold for rejecting possible trajectories provided the most human-like behavior in our model. For each pose in the predicted future trajectories, we compute the risk along the highway using the rectangular higher-order Gaussian, yielding 25 individual risk cost maps spanning a 5-second planning horizon. For each individual vehicle's risk field, the risk is scaled by its probability to reflect that trajectories less likely to occur are considered safer. We define the risk threshold as the maximum cost the lane change planner can incur in one action, or the maximum risk the vehicle is allowed to experience at one timestep. This parameter is defined separately for each simulated vehicle, as varying it creates different driving behaviors that are more or less aggressive.

After generating these risk fields, we implement a simple lane change planner by discretizing the risk fields into a graph, with edges for continuing in the lane, and for changing lanes. For each edge in the graph, we assume that the ego vehicle continues at its current velocity, and select the risk cost map corresponding to the appropriate time, then weight that edge of the graph as the risk at the position on the road. Using Dijkstra's algorithm, we can plan a simple least-cost path along the planning horizon that represents the lane change maneuvers the vehicle should perform while traveling along the highway. With this optimal path planned, we control the vehicle along the path using a simple MPC steering controller. However, the controller will only make a lane change when doing so does not exceed the risk threshold. After taking one control step, the planning step is run once again based on updates in the environment. This means that the planned path changes regularly as surrounding vehicles interact with the ego vehicle.

B. Learned Parameters

TABLE I
LEARNED PARAMETERS FOR MODEL

	Lane 1 (Slowest)	**Lane 2**	**Lane 3**	**Lane 4**	**Lane 5 (Fastest)**
1 σ slower	0.4224	0.5030	0.5169	0.4994	0.4460
Mean velocity	0.4017	0.4868	0.4862	0.4677	0.4386
1 σ faster	0.4123	0.4962	0.5070	0.4891	0.4421

To learn parameters as additional costs for the risk-aware planning, we divide the NGSIM data from I-80 and US-101 in half, reserving half of the data to validate our model. We take the learning portion of our data and filter it to find the smoothed positions and velocities of all the vehicles. We also extract the current lanes of each vehicle from its position to accurately identify each vehicle's lane.

For each timestep of each vehicle in the data, we compute the risk field at the current timestep using the rectangular cost function, and record the risk the vehicle is experiencing, along with the velocity, current lane, and average velocity of all vehicles in the current vehicle's lane. Using these data, we can apply an additional cost to the simulated vehicles based on their desired velocity, identifying which lane would be the safest to travel at the desired speed and penalizing all other lanes by a factor. The data can be found in Table I for cars traveling at the mean velocity, and for cars traveling one standard deviation above and below the mean. A visualization of these data can be seen in Figure 1. These data are then fed back into the cost function of the planner to tune the risk-aware planner to behave like a human driver in its lane selection.

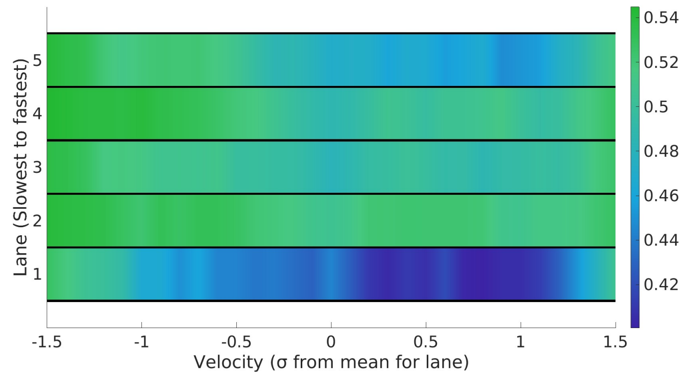

Fig. 1. A visualization of the risk in each lane relative to the velocity of the vehicle. This is a visualization of data seen in Table I.

V. SIMULATION AND DISCUSSION

A. Model validation

In this section, we validate our model using real-world data and show specific examples of our framework generating human-like behavior not found in other simulation methods for human drivers. We validate the model by using the half of the NGSIM data that we did not use to learn the parameters, and identify all the lane change maneuvers in the data. This leaves us with approximately 1,000 individual lane change maneuvers to validate the data.

Further, we subdivide the data by examining the lateral velocities of each driver during their maneuvers and categorizing the drivers as conservative, average, or aggressive. We define drivers with the lower quartile of lateral and longitudinal accelerations during lane changes as conservative, drivers in the upper quartile as aggressive, and the rest as average. We define three different parameters for aggressiveness that represent the varying driving styles, and validate each lane change maneuver with the corresponding model.

To perform the validation, we replace one of the recorded vehicles with our model 2 seconds before the lane change occurs in the data, and observe the planned trajectory for the next 4 seconds to determine if the model changes lanes into the same lane as the actual data. We also record the trajectory errors as both the final and average deviation from the planned path compared to the actual one. The results are given in Table II.

These results indicate an ability of our model to predict human actions in lane-changing maneuvers. While the framework is not specifically designed as a trajectory prediction method, its ability to predict the trajectories of vehicles suggests that

TABLE II
RESULTS OF DRIVER MODEL VALIDATION

	Aggressive Driver	Average Driver	Conservative Driver
Lane changes identified	75.06%	79.32%	82.41%
Average RMSE	4.88 m	4.21 m	4.25 m
Final RMSE	6.97 m	6.45 m	6.71 m

it effectively replicates human driving behavior, thus demonstrating its capacity to capture human behavior. The trajectory error is comparable to other methods of trajectory prediction in highway scenarios.

B. Example Cases

To demonstrate the ability of this model to generate new human behaviors we present the following two scenarios.

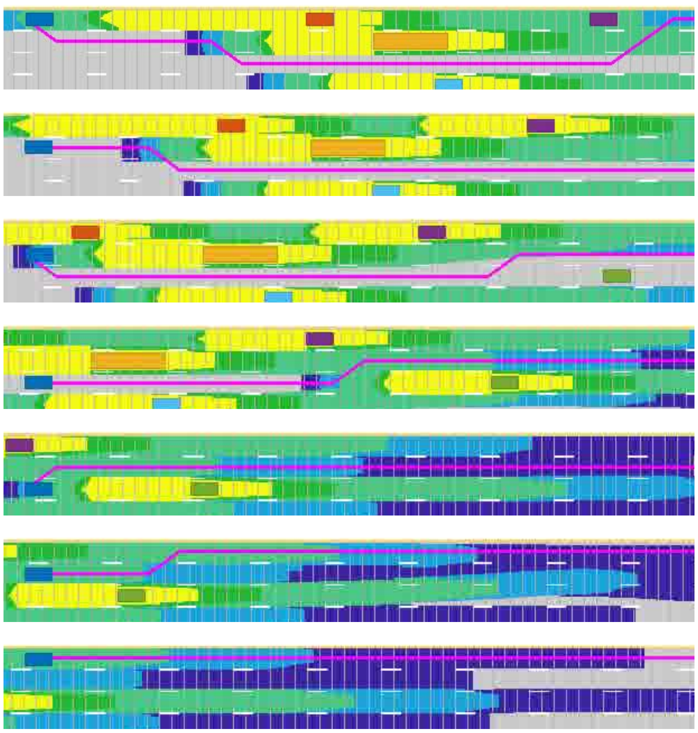

Fig. 2. Sequential timesteps of the simulation demonstrating the models ability to weave through traffic. The ego vehicle pictured in dark blue can be seen starting behind all vehicles along the road but with a higher velocity than all surrounding vehicles. As such the risk-aware planner navigates a safe route shown in purple to weave through the traffic that the ego vehicle executes. As the ego vehicle travels along the highway it regularly re-plans its trajectory as the predicted trajectories of surrounding vehicles update. Video

1) Weaving through traffic: The images in Figure 2 are taken at sequential timesteps from the simulator. The ego vehicle, depicted in dark blue, has a higher desired velocity and exhibits more aggressive behavior compared to the surrounding vehicles. As such, it weaves through the traffic, representing the high-level human behavior of aggressive drivers. This example demonstrates the effectiveness of our framework at generating various kinds of behaviors. Many current human behavior models struggle to weave through dense traffic, as the ego vehicle cannot reach a higher velocity within the short planning horizon of many models. This clearly shows the advanced capabilities of our model.

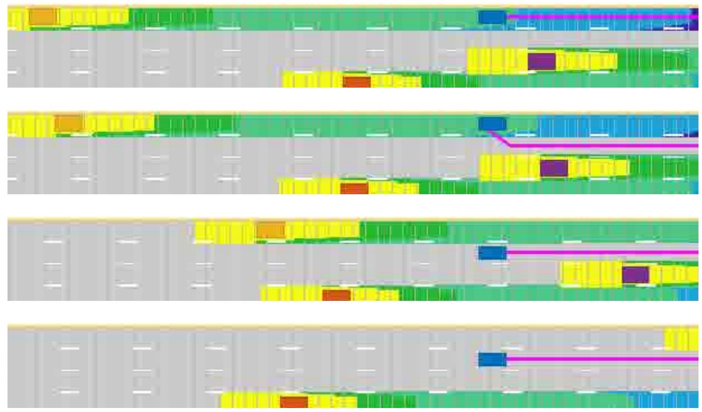

Fig. 3. Sequential timesteps of the simulation demonstrating the models ability move out of the way of an oncoming high-speed vehicle to maintain safety. The ego vehicle pictured in dark blue can be seen starting in front of a high speed vehicle and traveling at a slower velocity than the oncoming vehicle. As such the risk-aware planner navigates a safe route shown in purple to move out of the way of the approaching vehicle that that the ego vehicle executes. As the ego vehicle travels along the highway it regularly re-plans its trajectory as the predicted trajectories of surrounding vehicles update. Video

2) Moving out of way of high-speed vehicle: The images in Figure 3 are also taken at sequential timesteps from the simulator. The ego vehicle, depicted in dark blue, has a lower desired velocity and is more conservative than the surrounding vehicles on the road. In this situation, a high-speed car is approaching the ego vehicle from behind. Using our Risk-Aware planning framework, the ego vehicle changes lanes to allow the faster approaching vehicle to pass. This behavior effectively models a conservative driver who avoids dangerous situations to minimize their own risk.

VI. CONCLUSION

In this paper, we propose the Hierarchical Learned Risk-Aware Planning Framework for modeling human drivers. We provide validation and results that demonstrate the effectiveness of this model in replicating human behavior. Moreover, specific examples of generated trajectories showcase the model's ability to produce human-like patterns in new contexts. For future work, we plan to enhance the model to account for different driving characteristics, such as distracted driving. We also aim to learn additional parameters to enable the model to better mimic human behavior. Furthermore, we intend to conduct additional validation of the model in varying contexts to prove that the proposed approach generalizes to any situation.

VII. References

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